

Taleb Lean v0.2.6

Independent audit | 20 September 2026 | Mathematics, verification and proof progress

Assessment

Accept the mathematics and T061 discharge. Correct the release documentation before the next handoff. Reverse-Weibull and the positive location/scale API complete the previously recorded construction scope. The three maximum-parameter rules are now equalities of actual probability measures. No defect was identified in the new Lean mathematics.

Independent reproduction passed: clean build, 149 public checks, all 270 project constants within the allowed axioms, all schema probes, and all 14 regression fixtures. The regression coverage used three isolated, disjoint invocations of the unchanged harness.

Area	Independent assessment
Mathematics	22 theorems, 5 instances and 7 definitions added. Support branches, shape restrictions, positive-scale CDF rules and maximum-law parameters are correct.
T061 scope decision	The original target names the three families separately. The unified GEV parameterization and shape-to-zero limit remain explicit source obligations; they are not a retroactive closure gate.
Verifier repairs	The shared schema validator rejects all three prior mutations. Namespace/section discovery is repaired; the independent environment scan remains the trust check.
Remaining findings	One medium-priority handoff inconsistency: the Markdown ledger still records T061 as partial. Two low-priority groups cover continuity prose and task/parameter routing.
Release identity	ZIP, manifest, bundled commit and both patch routes agree. Dependency pins and existing mathematical statements/proofs are preserved.

Broader progress

2 discharged / 9 partial / 4 reuse / 91 missing / 34 source-check / 15 model-needed / 3 empirical = 158 families. T061 joins T060 as the second discharged family. All four planned T061 children are now complete. The other ten supported families retain their prior statuses.

There are 107 family/declaration citation pairs over 106 distinct names, up from 83/82. Declaration counts are implementation inventory, not independent book theorems. The 158 unequal families are a curated working ledger, not an exhaustive atomic census or a measure of effort completed.

Mathematics and closure boundary

The reverse-Weibull construction and the positive affine API are sound

Source: §9.1, printed p.173 / PDF p.187. Reverse-Weibull is the upper-endpoint family $W\alpha(x) = \exp(-(-x)^\alpha)$ for $x < 0$ and 1 for $x \geq 0$, with $\alpha > 0$ and book extreme-value index $-1/\alpha$. This is distinct from the positive-support Weibull lifetime law.

Obligation	Reviewed result
Actual probability law	Monotonicity, right continuity and endpoint limits 0/1 give a Stieltjes probability measure. <code>cdf_reverseWeibullMeasure_apply</code> identifies its CDF globally, including $x = 0$.
Right continuity for any α	Correct: at a negative point the power base is positive; at and above zero, the function restricted to the right neighborhood is constant 1. Positive α is still required for the probability-law constructor.
Analytic $n = 0$ identity	$W\alpha(x)^n = W\alpha(n^{1/\alpha}x)$ is valid for every natural n . At $n = 0$ both sides are 1. Actual maximum-law theorems retain finite and nonempty sample hypotheses.
Affine transformation	<code>affineLaw v μ σ</code> is the measurable pushforward by $\mu + \sigma z$. For $\sigma > 0$, its CDF is <code>cdf v ((x - μ)/σ)</code> . The exact <code>lic</code> preimage avoids any boundary-mass shortcut.
Realization and equality	Standard-law product models exhibit iid coordinates. The three affine maximum theorems use <code>Measure.eq_of_cdf</code> with real probability instances on both sides. Their ordinary hypotheses are not vacuous.

Coordinates, scale $\sigma > 0$	Maximum of $n > 0$ iid coordinates
Gumbel (μ, σ)	Location $\mu + \sigma \log n$; scale σ .
Fréchet $(\xi > 0, \mu, \sigma)$	Lower endpoint μ ; scale σn^ξ .
Reverse-Weibull $(\alpha > 0, \mu, \sigma)$	Upper endpoint μ ; scale $\sigma n^{(-1/\alpha)}$.

These are the endpoint coordinates of the three displayed book families. They must not be silently identified with the location parameter of the unified GEV kernel. The affine definitions and probability instances themselves permit any real σ ; the CDF and maximum rules correctly require $\sigma > 0$.

T061 meets its preserved target: construct global Gumbel, Fréchet and reverse-Weibull CDFs and prove validity, including the recorded positive location/scale child. Densities, domains of attraction, the unified GEV parameterization and its $\xi \rightarrow 0$ limit are not proved here. The last two are visible in the source and should remain explicitly tracked outside the completed scope.

Verifier and previous findings

A schema gate checks structure; the Lean environment checks declarations and trust

V025-L1 is repaired

One `backlog_schema.validate` function now serves the generator and verifier. It requires the exact T001...T158 sequence, typed fields, known vocabularies, no unknown fields, joint presence of delivery fields, and agreement between discharged status, state and Boolean. It rejects the prior duplicate-ID, false discharged flag and numeric-scope mutations.

The shipped schema suite has 12 deliberately malformed ledgers plus one valid control: 12/12 rejected, 13/13 expected outcomes. Those two reporting conventions are consistent. Independent Python probes additionally check the generator refusal before writing, discovery behavior and failure paths; they are separate from real Lean execution.

H026-1 is repaired and disclosed accurately

The old regex discovery popped a namespace at a section end. The replacement stack tracks namespace and section scopes together, adding only namespaces to declaration names. The new positive control places declarations after section boundaries inside a namespace; the shipped two-section mathematics modules exercise this pattern too.

The defect would have blocked release through stale or unknown names. It was not evidence that an invalid proof had passed. The regex remains deliberately limited; agreement with the environment inventory catches missed public theorem/instance declarations, while the separate trust scan checks every project constant before visibility filtering.

Earlier documentation repairs

The extrema bridge now uses the exact boundary events $\{\min_{RV} X = x\}$ and $\{\max_{RV} X = x\}$. Replacement-map construction clauses are correctly historical, and the audit-history response names both Gumbel and Fréchet. T061 carries `law_theorem` in the JSON ledger. The current Markdown ledger was missed, as detailed on the next page.

What remains a review judgment

Schema validity and declaration existence do not prove that a declaration meets its family target. The current all-constant scan, recursive module coverage, alias checks, dependency pins and clean-worktree gates remain intact. A few deeply malformed Python values can raise a caught exception instead of a tailored schema diagnostic; that fails closed and is not an acceptance bypass.

No new proof-trust or verifier-acceptance blocker was found. The remaining generation gap is that the human-readable ledger is not regenerated or checked by the shared JSON schema workflow.

Required handoff corrections

Three consolidated findings; none requires changing the new mathematical proofs

D026-1 | medium | Markdown ledger is a full release behind

docs/FORMALIZATION_BACKLOG.md is byte-identical to v0.2.5. Its T061 row says partial, lists 21 old credited declarations and leaves reverse-Weibull/affine work open. JSON correctly says discharged with 45 names. Seven rendered T061 fields differ. An independent copy-run confirms the generator reproduces JSON but leaves Markdown unchanged.

Repair: render the Markdown rows from the same validated records, and add a fast generation-consistency check for both formats. Do not change T061 back to partial. Also update the ExtremeValueLaws.lean module title and paragraph that still call the reverse-Weibull and location/scale children unfinished; point the latter to ExtremeValueAffine.lean.

D026-2 | low | false continuity description

T061 JSON and its generator say the Fréchet and reverse-Weibull CDFs are not globally continuous, only right-continuous. Under the admitted positive shapes, the support-correct piecewise CDFs are globally continuous. Fréchet tends to 0 from above at its lower endpoint; reverse-Weibull tends to 1 from below at its upper endpoint.

Repair: say that right continuity and endpoint limits are formally proved and sufficient for the Stieltjes construction. A separate global-continuity theorem is not exported or needed for closure. Keep the totaled-power warning attached to the unguarded frechetFormula, not to the repaired frechetCDF. No extra proof is requested.

D026-3 | low | task routing and reciprocal shape

The handoff routes EVT densities to T062, but T062 actually targets regular-variation-to-Fréchet convergence and Gaussian Gumbel normalizers. This is a carried-forward documentation error, including earlier audit guidance. Preserve T062 and put density work in a separately identified source task; T011 concerns the general convergence/domain formulation.

FABLE_REVIEW §16.6 also proposes Pareto maxima converging to frechetMeasure α . If α is the Pareto tail exponent, this project requires frechetMeasure $(1/\alpha)$, because its shape ξ gives CDF $\exp(-x^{-(1/\xi)})$. For Pareto lower bound L , the natural normalization is $M_n/(L n^{(1/\alpha)})$, yielding $\exp(-x^{-(\alpha)})$ for $x > 0$.

Repair: correct those references in the authoritative generator, ledger and current handoff. Keep the unified GEV parameterization and $\xi \rightarrow 0$ limit visible as outstanding source tasks without reopening T061 or claiming they are already covered by another ID. Record any newly created tasks separately until the parent ledger is deliberately revised.

Position in the broader ledger

Two closed families; nine supported families remain partial

Family	Status	Audited delivery scope
T001	Partial	Ratio-only regular/slow variation API; positive measurable convention and its bridge remain.
T007	Partial	Gaussian realization at stable index 2, including zero scale; general stable existence remains.
T008	Partial	Subexponential predicate and conditional self-convolution tail result; concrete-law examples remain.
T029	Partial	Two-power-tail formula exponent; random-variable weighted-sum squeeze remains.
T032	Partial	Analytic power-transform exponent; actual law pushforward remains.
T046	Partial	Actual Gaussian characteristic-function and convolution bridge; Cauchy remains open.
T047	Partial	Prerequisite-only sum/convolution infrastructure; average-law scaling and convergence remain.
T060	Discharged	Generic maximum CDF and minimum survival laws, with ordinary probability hypotheses.
T061	Discharged	All three standard probability laws, exact CDFs, affine families and exact maximum-law parameters.
T118	Partial	Prerequisite-only tail-kernel convexity; Pareto moments and integrated Jensen remain.
T124	Partial	Rational-expression algebra and positivity; gamma inverse-moment integration remains.

The discharged count rises from 1 to 2; partial falls from 10 to 9. The supported-family count remains 11. T061 gains 24 citation pairs, taking its own list from 21 to 45. The shared Gaussian declaration explains why 107 family/name pairs cover 106 distinct names.

There are **140 formal-proof families**: 2 discharged and 138 still open, comprising 9 partial, 4 reuse, 91 missing and 34 source-check. Another 15 require models and 3 are empirical. This remains an early stage of the broader formalization, with a now-complete EVT construction component to build on.

The companion progress JSON preserves all original 158 rows and adds the eleven audited scopes, dependency-group counts, completed T061 children and explicit supplemental obligations. The Markdown gives a readable full index. Unchanged unsupported rows are tallied, not represented as independently re-audited mathematics.

None of these counts measures upstream Mathlib integration, numerical execution or readiness of the separate quantum/classical application. No release evidence establishes those software milestones, and no percentage of total effort can be inferred from the parent-family tally.

Next milestone and acceptance scope

Repair the handoff, then implement T029 before expanding the Pareto slice

1. T029: actual nonnegative weighted sums

For measurable nonnegative X and Y , positive weights a and b , and finite log-tail exponents α and β , target the exponent $\min(\alpha, \beta)$ for $aX + bY$. State the survival functions as probabilities of strict tail events. Independence is unnecessary; almost-sure nonnegativity is sufficient if the event arguments account for null sets.

For $Z = aX + bY$ and positive t , use $P(X > t/a) \leq P(Z > t)$, $P(Y > t/b) \leq P(Z > t)$, and $P(Z > t) \leq P(X > t/(2a)) + P(Y > t/(2b))$. Prove positive argument-rescaling and the finite-log-exponent squeeze/sum facts, with eventual positivity and log-denominator sign stated explicitly.

The existing `two_power_tail_min` is an exact power-formula result; it does not by itself cover arbitrary tails with the same finite exponent. Finish the general analytic bridge, then the probability statement. Distinguish exponent equality from survival-ratio asymptotics or regular variation of the sum. A finite-family extension can follow without adding independence.

2. Exact Pareto against the pinned probability law

Use a positive lower endpoint L and shape $\alpha > 0$. Identify the global CDF/survival, then compute moments with the exact domain: for $p \geq 0$, the moment is finite exactly when $p < \alpha$ and equals $\alpha L^p / (\alpha - p)$; at and above α , prove divergence in an extended nonnegative integral interface. Do not use a totalized real integral as evidence of a finite moment.

Then prove threshold conditioning/excess and the positive-power pushforward. Credit only the delivered slices of T005/T021/T028/T032; their parent targets include more than a single Pareto example. The `affineLaw` API supplies measurable affine pushforwards, not the integration or conditioning arguments for free.

3. Later convergence and density work

Pareto maxima are a contained first domain-of-attraction example under T011/T062, with the reciprocal Fréchet shape corrected. Gaussian normalization is a separate T062 obligation. EVT densities and `withDensity` identifications need their own correctly tracked source scope. Unified GEV and the $\xi \rightarrow 0$ Gumbel limit remain additional source work, not silently completed consequences of this release.

The accompanying Fable handoff lists concrete edits and acceptance checks. The received release has been preserved unchanged; this audit supplies findings and evidence, not an untested modification of the proof project.

Independent execution

Exact commands and complete logs accompany the audit package

Check	Observed outcome
Clean build	PASS; exit 0, 106.49 s; 2743 Lake jobs and no compiler warnings/errors.
Normal verifier	PASS; exit 0, 217.96 s; 125 theorems, 8 instances, 16 aliases.
Trust and ledger	270 constants, 69 internal; only propext, Classical.choice, Quot.sound. 158 families; 107 citations over 106 names.
Schema probes	PASS; all 12 malformed cases rejected and the valid control accepted.
Regression coverage	PASS 14/14, exactly once; 20.9 min elapsed across three concurrent, isolated harness invocations. All restoration checks pass.
Final source/dependencies	All 196 received files outside generated evidence match byte for byte. All nine dependency revisions match and worktrees are clean.

Official Lean 4.24.0 and all nine exact pinned dependencies were restored after the previous temporary cache was removed. The project build began without project build outputs. All three unchanged harness commands used relative scratch/output arguments and distinct copies. Both private-sorry fixtures also exposed sorryAx in standalone scans. This is partitioned full fixture coverage, not one sequential invocation; complete commands and individual results are retained.

Fixture	Required behavior
valid	Accept the pristine project.
namespace_section_theorem	Accept correctly qualified declarations after section ends.
nested_imported_module	Accept and scan the imported nested module and theorem.
private_sorry_theorem / private_sorry_def	Reject; standalone scan must also expose sorryAx.
private_axiom	Reject the added project axiom.
structure_namespace_theorem / protected_theorem	Reject a public-inventory mismatch.
alias_map_mismatch / optimized variant	Reject alias mismatch, including Python -O.
orphan_module / nested_orphan_module	Reject unimported project modules.
dirty_dependency	Reject a modified dependency checkout.
backlog_duplicate_id	Reject the malformed ledger through the real verifier.

Fourteen fixture names are grouped above for space. A negative fixture succeeds only when it rejects for the expected reason. Source-restoration hashes and build-snapshot resets are checked separately. No received proof statement, allowed-axiom policy or fixture was changed for the reproduction.

Provenance and evidence boundaries

The audit separates archive identity, supplied claims and observed execution

Item	Independently checked
Release ZIP	SHA-256 d684d0ef3c3d17f9ceb5cd456969559674c1ac86e21bbd6a27505cf22d4fb3f3; 205 regular files.
Manifest and paths	204/204 SHA256SUMS entries; no unsafe paths, duplicates, symlinks or included project build artifacts.
Bundled commit	83ef501d87a1c910cc166f880ed44b7094e6f5d3; every packaged file equals its tracked counterpart.
Both patch routes	The full v0.2.0 route and incremental v0.2.5 route apply to the stated bases and reproduce the same packaged tree.
Pins and preservation	Lean 4.24.0; Mathlib f897ebcf72cd16f89ab4577d0c826cd14afaafc7. Existing mathematics preserved; extrema bridge edits are documentation only.
Source correspondence	Locally supplied arXiv:2001.10488v4 PDF, 523 pages. Maximum law: Eq. (9.1), printed172/PDF186. Named EVT forms and shape conventions: printed173/PDF187.

All 15 retained-artifact checksums pass. The newly retained v0.2.5 audit PDF, progress Markdown and evidence ZIP are byte-identical to the prior independent outputs.

History is narrower evidence than a live worktree

The bundle proves its included history and archive equality. It does not establish the author's original worktree cleanliness, unpushed state or a later delivery commit. It contains three commits after the v0.2.5 deliverables commit, or four after the v0.2.5 packaged commit; an additional v0.2.6 deliverables commit is outside the advertised bundled scope.

What this audit does and does not establish

The evidence supports the specific Lean propositions, their permitted axiom dependencies, reviewed correspondence to the source formulas and the stated family-level progress. It does not certify every mathematical claim in the book, an upstream Mathlib contribution, empirical claims, computational performance or deployment of the separate application.

The uploaded validation log is corroborating release evidence. The independent execution record on the previous page identifies exactly what was rerun. The Python-only schema and generation probes are labeled separately and must not be counted as additional proof checks.

The audit package contains the PDF and editable report, the full progress ledger, the Fable handoff, source subreviews, bounded probe scripts/results, provenance checks and execution logs. Compiler binaries, dependency caches, source-book bytes and duplicate submitted archives are excluded. SHA256SUMS covers the included evidence files.